Project2_Launch_JansenRinck

September 22, 2026

1 Project 2: Launch

1.1 Research Question:

Between 2018 and 2020, does the rate at which lightning strikes convert into wildfire ignitions vary by fuel type across the ecoregions of the contiguous United States, independent of raw lightning exposure?

2 Data:

1. Spatial wildfire occurrence data for the United States, 1992-2020 (6th Edition)(USFS)
   - Federal, State and Local fire data aggreagated
   - Discovery date, final fire size, point location (1 sq mile accuracy), wildfire-cause
   - 2.3 million geo-referenced wildfire records
2. Combined Geostationary Lightning Mapper (GLM) Hourly Thunder Hour Data Product (NASA)
   - 2019 - 2025
   - Two sattelites GOES-18 (west) and GOES-19 (east)
   - Thunder hour: an hour with detected lightning at a location - a frequency-based storm-activity metric, less biased by a sensor's geographic detection-sensitivity variation than raw flash counts
   - Resolution: 5 km
3. LANDFIRE's 40 Scott and Burgan Fire Behavior Fuel Model (FBFM40), CONUS raster (USFS & USGS
   - 40 fuel models classifying surface fuel loading by type (grass, brush, timber, slash), size class, bed depth, and moisture of extinction
   - Fuel assignments aren't static - herbaceous fuels shift between live and dead through the growing season, and burned or otherwise disturbed areas are updated to reflect estimated recovery through 2022
4. Level III Ecoregions of the Conterminous United States (EPA)
   - Ecoregions mark areas of broadly similar ecosystems - a spatial framework for managing and comparing environmental resources across agencies
   - Delineated from patterns across multiple factors (geology, vegetation, climate, soils, land use, wildlife, hydrology), weighted differently region to region
   - Four Levels: Level I (15 regions, North America) → Level II (50) → Level III (85 in the conterminous US) → Level IV (further subdivisions)
5. Backup Wildfire Data: Interagency Fire Occurrence Reporting Modules (InFORM
   - Ongling incorporation of historic data across other agencies is still in progress
   - Data still needs to be backfilled

2.1 Data Processing:

1. CRS
   - Albers Equal Area Conic
     - Preserves area calculation accuracy
     - Minimal distortion for CONUS
2. Zonal Statistics at CONUS Scale for LANDFIRE Raster
   - Windowed (chunked) reads for the raster data set to avoid memory issues in Docker
3. Lightning Data Cropping to CONUS

4. LANDFIRE Attribute Table
    - Translate values to fuel model classes
5. Categorize Fuel Types
    - Single majority class per eco region?
    - Proportions of fuel class per eco region?
    - Lightning strikes per fuel class

2.2 Analysis Methods:

2.2.1 Spatial joins and overlays

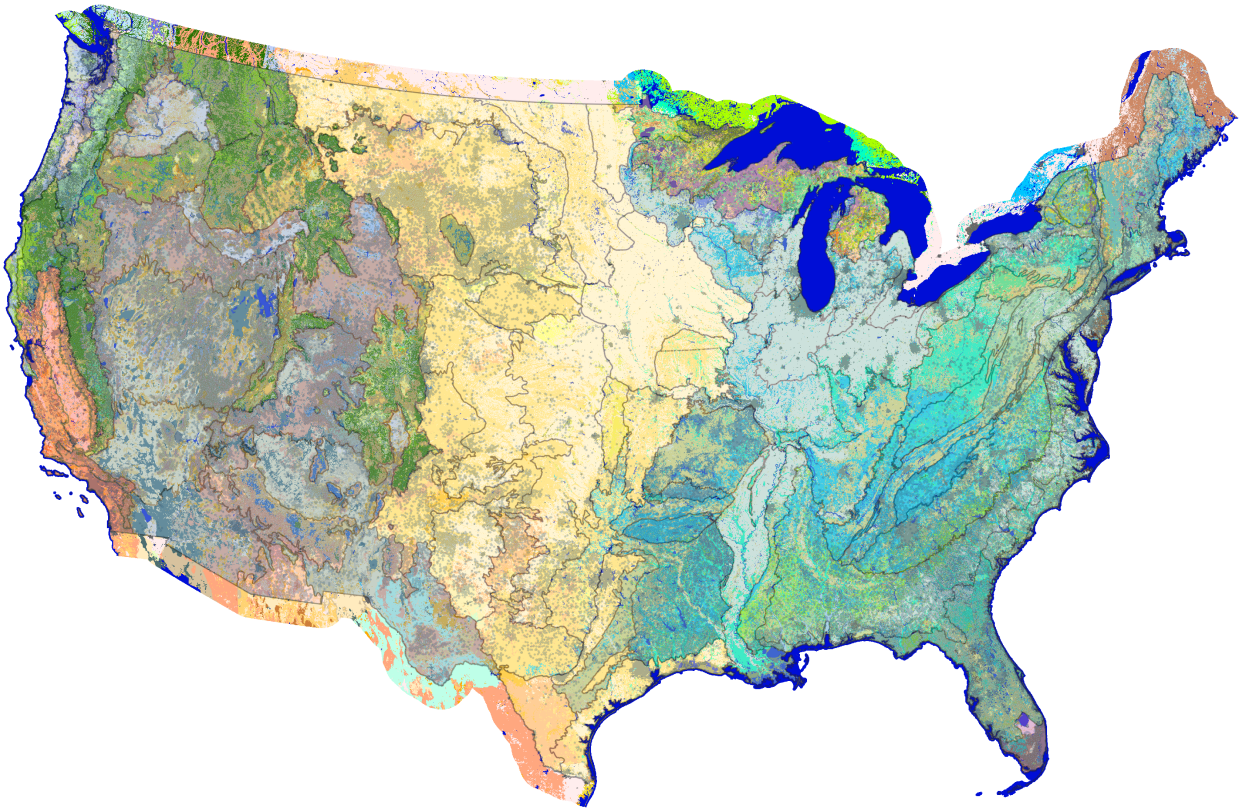
- Fire points joined to ecoregion polygons (point-in-polygon)
- Fire points sampled against the fuel raster (tagging each ignition with its fuel class)
- Lightning grid cells overlaid on the fuel raster to assign each cell a majority fuel class
- Statistics of the fuel raster within ecoregion polygons (raster clipped by polygon)

2.2.2 Attribute filtering and aggregation

- Filtering fire records to NWCG_CAUSE == "Natural" and to the date range February 2019–December 2020
- Lightning will use the combined Monthly Thunder Hour product for 2019 and 2020 to summ across CONUS to get a per grid cell count
    - Monthly Thunder Hour: did lightning happen in this cell during this hour, yes or no?
- Aggregating fuel pixel counts within each ecoregion into classes

2.2.3 Choropleth mapping and advanced visualization

- Fuel map colored by their ignition-to-thunder-hour ratio
- Ecoregions colored by their ignition-to-thunder-hour ratio
- Create a map that displays the areas with the most lightning cause wildfire ignitions



[1]:
```
# %pip install earthaccess
# import earthaccess
#import glob

#files = sorted(glob.glob("data/glmmth/GLMc_th_*.nc"))
# earthaccess.login(strategy="interactive", persist=True)

# results = earthaccess.search_data(
#     short_name="glmmth",
#     temporal=("2019-01-01", "2020-12-31"),
# )
# files = earthaccess.download(results, "work/Project2/data")
```

[2]:
```
# For each file, promotes lat/lon to real coordinates and crops to CONUS, then sums each␣
 ↪year's 12 monthly slices and adds both years together into
# one cumulative thunder-hours grid

# Gridded because "thunder hour" is a binning concept by definition, and because it merges␣
 ↪detections from instruments with different viewing geometries
# (both satellites, and more broadly ground networks) onto one common reference frame

# lat is 520 values, lon is 1180 values, and total is 1180×520 - like a spreadsheet: lat␣
 ↪gives the row headers, lon gives the column headers, and cell
# total[i, j] is the thunder-hour count

import glob
```

```
import xarray as xr

files = sorted(glob.glob("data/glmmth/GLMc_th_*.nc"))

def load_and_crop(path):
    ds = xr.open_dataset(path)
    # lat/lon are plain data variables here, not indexed coordinates -
    # promote them so .sel() can crop by actual lat/lon values
    ds = ds.assign_coords(lat=("nlat", ds.lat.values), lon=("nlon", ds.lon.values))
    ds = ds.swap_dims({"nlat": "lat", "nlon": "lon"})

    # handle either latitude ordering (ascending or descending)
    lat_vals = ds.lat.values
    lat_slice = slice(24, 50) if lat_vals[0] < lat_vals[-1] else slice(50, 24)

    return ds.sel(lat=lat_slice, lon=slice(-125, -66))

datasets = [load_and_crop(f) for f in files]

# If range assumptions above don't match this file and need adjusting
print("Cropped shape:", datasets[0].thunder_hours.shape)

# sum each file's 12 months, then sum both years together
total = sum(ds.thunder_hours.sum(dim="nmon") for ds in datasets)

print(total)
```

```
Cropped shape: (1180, 520, 12)
<xarray.DataArray 'thunder_hours' (lon: 1180, lat: 520)>
array([[ 11,  10,   9, …,   0,   0,   0],
       [ 16,  14,  12, …,   0,   0,   0],
       [ 21,  17,  16, …,   0,   0,   0],
       …,
       [ 14,  16,  17, …, 184, 181, 182],
       [ 19,  17,  16, …, 180, 185, 197],
       [ 20,  18,  16, …, 183, 191, 195]])
Coordinates:
  * lat      (lat) float32 49.97 49.92 49.88 49.83 … 24.17 24.12 24.08 24.02
  * lon      (lon) float32 -125.0 -124.9 -124.9 -124.8 … -66.12 -66.07 -66.03
```

[3]:
```
# counts of each fuel class per polygon
import os
import geopandas as gpd
import pandas as pd

cache_path = "data/processed/fire_natural_2019_2020.gpkg"

if os.path.exists(cache_path):
    fire = gpd.read_file(cache_path)
else:
    fire = gpd.read_file("data/processed/fire_occurrence_albers.gpkg")
```

```
    fire["DISCOVERY_DATE"] = pd.to_datetime(fire["DISCOVERY_DATE"])
    fire = fire[
        (fire["NWCG_CAUSE_CLASSIFICATION"] == "Natural")
        & (fire["DISCOVERY_DATE"] >= "2019-02-12")
        & (fire["DISCOVERY_DATE"] <= "2020-12-31")
    ]

    fire.to_file(cache_path, driver="GPKG")
```

[4]:
```
print(fire.head())
```

```
         OBJECTID     FOD_ID      FPA_ID SOURCE_SYSTEM_TYPE SOURCE_SYSTEM  \
2166748   2166749  400500002  FS-6955822                FED   FS-FIRESTAT
2166749   2166750  400500003  FS-6936597                FED   FS-FIRESTAT
2166751   2166752  400500005  FS-6955006                FED   FS-FIRESTAT
2166752   2166753  400500006  FS-6947814                FED   FS-FIRESTAT
2166753   2166754  400500007  FS-6957509                FED   FS-FIRESTAT

        NWCG_REPORTING_AGENCY NWCG_REPORTING_UNIT_ID  \
2166748                    FS                USMTKNF
2166749                    FS                USMTKNF
2166751                    FS                USMTKNF
2166752                    FS                USMTKNF
2166753                    FS                USMTHLF

                     NWCG_REPORTING_UNIT_NAME SOURCE_REPORTING_UNIT  \
2166748                  Kootenai National Forest                 0114
2166749                  Kootenai National Forest                 0114
2166751                  Kootenai National Forest                 0114
2166752                  Kootenai National Forest                 0114
2166753  Helena - Lewis and Clark National Forest                 0115

                     SOURCE_REPORTING_UNIT_NAME  … FIRE_SIZE  \
2166748                  Kootenai National Forest  …       0.1
2166749                  Kootenai National Forest  …       0.6
2166751                  Kootenai National Forest  …       0.1
2166752                  Kootenai National Forest  …       0.1
2166753  Helena - Lewis and Clark National Forest  …       0.1

        FIRE_SIZE_CLASS   LATITUDE   LONGITUDE OWNER_DESCR STATE COUNTY  \
2166748               A  48.882222 -115.020556     PRIVATE    MT    053
2166749               B  48.484167 -115.258056        USFS    MT    053
2166751               A  48.457500 -115.913333        USFS    MT    053
2166752               A  48.647778 -114.830556        USFS    MT    029
2166753               A  46.913333 -112.488889        USFS    MT    049

        FIPS_CODE              FIPS_NAME                          geometry
2166748     30053          Lincoln County  POINT (-1402356.177 3016090.463)
2166749     30053          Lincoln County  POINT (-1428438.760 2976734.126)
2166751     30053          Lincoln County  POINT (-1476952.178 2983877.638)
2166752     30029         Flathead County  POINT (-1393606.561 2988029.143)
2166753     30049  Lewis and Clark County  POINT (-1255217.511 2767266.359)
```

```
[5 rows x 39 columns]
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[4], line 2
      1 print(fire.head())
----> 2 print(ecoregions.head())

NameError: name 'ecoregions' is not defined
```